Short Notes on Massless Gravitational Pressure Waves

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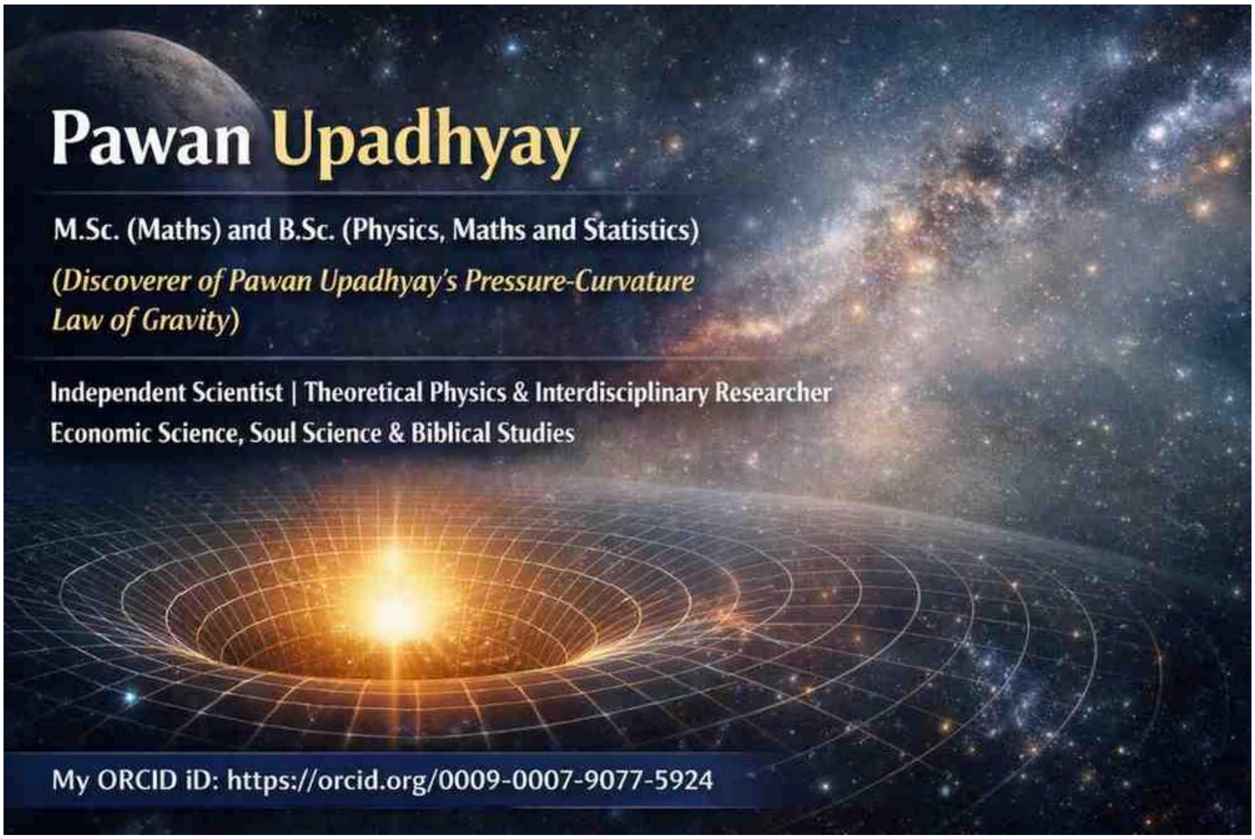

Short Research Notes on Massless Gravitational Pressure Waves

1. Light has the fastest known speed because the photon is a massless particle.

2. For massive particles:

$v < c$

and as energy increases,

$v \to c$

where the symbol "→" means "approaches" or "tends toward."

3. A photon is a special case because it is massless and travels exactly at the speed of light:

   $v = c$

4. In the PPC framework, gravitational pressure disturbances may propagate as gravitational pressure waves.

5. Gravitational pressure waves are massless, they travel at the speed of light:

   $v_{gpw} = c$
6. In the PPC model, gravitational waves are interpreted as gravitational pressure waves (GPWs). $v_{gpw} = v_{gw}$
7. The relation

   $m = 0 \Rightarrow v = c$

   expresses the principle that massless waves propagate at the invariant speed c.

8. Massless gravitational pressure waves are proposed to act as carriers of gravitational influence through spacetime, analogous to the role of electromagnetic waves in transmitting electromagnetic influence. Since these waves are massless, they propagate at the invariant speed of light c. In the PPC model, observed gravitational waves are interpreted as manifestations of such gravitational pressure waves.

9. Therefore, in the PPC model, the propagation speed of massless gravitational pressure waves is proposed to be equal to the speed of light.

Conclusion:

Massless gravitational pressure waves travel at the speed of light (c). Massive objects can only approach c, whereas truly massless entities propagate at c itself.

[Note: Points 1–3 are consistent with standard relativity.]

Massless Gravitational Pressure Waves

Light travels at the fastest known speed because the photon is a massless particle. Massive particles can only approach the speed of light, whereas massless particles travel exactly at the speed of light.

For massive particles:

$v \to c$

where the symbol "→" means "approaches."

For massless particles:

$v = c$

In the PPC model, gravitational waves are interpreted as gravitational pressure waves (GPWs). Since these waves are massless, they propagate at the speed of light:

$v_{GPW} = c$

Observations of gravitational waves show that their speed is equal to the speed of light within experimental accuracy. This is consistent with the interpretation that gravitational waves are massless gravitational pressure waves.

Therefore, the PPC model proposes that gravitational pressure waves are massless carriers of gravitational influence that propagate through spacetime at the invariant speed c.

> **Massless gravitational pressure waves travel at the speed of light.**

Symbolically,

$$m = 0 \quad \Rightarrow \quad v = c$$

or

$$v_{\mathrm{gpw}} = c$$

where v_{gpw} denotes the speed of the gravitational pressure wave.

The key point is that **masslessness implies propagation at** c in relativistic physics.

> **Light has the fastest speed because photons are massless. For photons, $v = c$. For massive particles, $v \to c$ but $v \neq c$.**

The observed speed of gravitational waves is **equal to the speed of light within experimental accuracy**.

A landmark observation was the 2017 event GW170817. Gravitational waves and electromagnetic radiation from the same neutron-star merger arrived at Earth almost simultaneously after traveling about 130 million years through space.

This observation showed that

$$v_{\mathrm{GW}} \approx c$$

where:

- v_{GW} = speed of gravitational waves,
- $c = 299,792,458 \ \mathrm{m/s}$ = speed of light.

Therefore, current observations are consistent with:

$$v_{\mathrm{GW}} = c$$

to a very high degree of precision.

[v_GPW = v_GW]